SANOJ K

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SUMMARY

Aeronautical Engineering graduate with hands-on 3D modeling experience in SolidWorks and CATIA V5, and applied FEA/CFD skills developed through structural, thermal, and damage-assessment simulations in ANSYS Fluent, SpaceClaim, Patran, and Nastran. Contributed to a fighter aircraft ECS project at the Aeronautical Development Agency (ADA), and gained shop-floor quality experience at HAL, including exposure to GD&T, non-conformities, and engineering change management. Brings a disciplined, standards-driven approach to design and validation.

TECHNICAL SKILLS

- 3D Modeling & Simulation: SolidWorks and CATIA V5 (hands-on across multiple projects); Finite Element Analysis in Patran and Nastran; ANSYS Fluent CFD and ANSYS SpaceClaim; structural and thermal load analysis.
- Structural Dynamics: Thermal-stress and high-temperature failure analysis on turbine blades; cockpit ECS thermal/acoustic optimisation against MIL-E-18927E military compliance standards; UAV structural weight and centre-of-gravity analysis in MATLAB.
- Quality, Standards & Documentation: GD&T, non-conformities and concessions, configuration management, engineering change management, tooling selection and validation, Standard Operating Instructions (SOI), M-BOM/E-BOM concepts.
- Manufacturing & DFM: Assembly process planning, repair routings, workshop support, and Design for Manufacturability, grounded in HAL shop-floor experience.

EXPERIENCE

Aeronautical Development Agency (ADA), Bengaluru, India Feb 2026 – May 2026
Aeronautical Engineering Intern

- Modeled baseline 3D geometry for a fighter aircraft Environmental Control System (ECS) in SolidWorks, validating designs against MIL-E-18927E thermal-performance requirements.
- Conducted ANSYS Fluent CFD simulations (k-ε model) to optimise cockpit thermal and acoustic performance across varying altitudes, and interpreted velocity and temperature contour data to inform design decisions.

Nexmentho (Aerospace Design Startup), Bengaluru, India Feb 2026 – May 2026
Aero-Structure Design Intern

- Performed Finite Element Analysis on structural components using Patran and Nastran to evaluate potential airframe damage and assess structural integrity under load.

Hindustan Aeronautics Limited (HAL) — Overhaul Division, Bengaluru, India Aug 2025 – Sep 2025
Aerospace Intern

- Studied aerospace maintenance, overhaul, and inspection procedures on the shop floor, developing a practical understanding of in-service repair methodology, non-conformities, and quality-validation gates.

Dautya Aerospace Pvt Ltd & HAL Foundry/Forge Division, Bengaluru, India Feb 2023 – May 2023
Engineering Intern

- Evaluated composite material applications and manufacturing constraints against aerospace quality-control practices, and produced accurate engineering documentation in collaboration with cross-functional teams.

PROJECTS

Military Cockpit ECS Analysis (MIL-E-18927E Standard) Internship Project

- Built a scaled 3D cockpit model in SolidWorks with optimised airflow inlets for thermal management.
- Processed geometry in SpaceClaim and ran Fluent CFD simulations (k-ε model) to map velocity and temperature contours.
- Validated thermal performance against military compliance standards to help ensure pilot safety.

Gas Turbine Blade Structural Analysis Academic Project

- Modeled 3D turbine blades in SolidWorks and performed thermal-stress and mechanical-load analyses in ANSYS to identify high-temperature failure points.

Foldable Wing VTOL UAV Stability Analysis Academic Project

- Designed 3D actuating wing mechanisms in SolidWorks and estimated structural weight and centre-of-gravity dynamics using MATLAB.

EDUCATION

Bachelor of Engineering in Aeronautics — CGPA 7.28 2023 – 2026
MVJ College of Engineering, Visvesvaraya Technological University, Bengaluru

Diploma in Aeronautics — CGPA 8.35 2021 – 2023
Hindustan Electronics Academy, Bengaluru

PUBLICATIONS & LEADERSHIP

- Publications: Co-authored (with team) a Taylor & Francis book chapter on computer vision-based robotics for healthcare, and an IJRASSET paper on UAV solutions for precision agriculture (Nov 2025).
- Team Leadership: President, UAV Club; Team Lead, NIDAR; Hardware Lead, Robotics Club.
- Event Coordination: Aeronautical Department Coordinator — organised a national-level technical fest.